Code No: 123AN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September - 2021
PROBABILITY AND STATISTICS
(Common to ME, CSE, IT, MCT, AME, MIE)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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Used Papers SEP 2021

1. A random variable X has the following probability distribution.

X:	0	1	2	3	4
P(X):	3k	3k	K	2k	6k

Find: a) k b) *var(X)* c) *var(2X-4)* d) $P(X<2)$ and e) $P(X>1)$. [15]

2.a) If 2% of the items of a factory are defective and the items are packed in boxes, find the probability that there will be atleast 3 defective items in a box of 100 items.

b) In a Normal distribution, 7% of the items are under 35 and 89% are under 63. Find the mean variance of the distribution. [7+8]

3. For the following bivariate probability distribution of X and Y.

X \ Y	0	1	2	3
0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{2}{32}$
1	$\frac{2}{32}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{32}$
2	$\frac{3}{32}$	$\frac{2}{16}$	0	$\frac{1}{16}$

Find: a) $P(X \le 1,\ Y = 3)$ b) $P(Y \le 2)$ c) $P(X \le 1, Y \le 2)$
d) $F_X(2)$ and e) $F_Y(1)$. [15]

4. Find the coefficient of correlation and the two lines of regression. [15]

X:	10	12	18	24	23	27
Y:	13	18	12	25	30	10

5. If samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6 without replacement, then find:
a) the mean and standard deviation of population.
b) the mean and standard deviation of the sampling distribution of means. [7+8]

6. Fit a Poisson distribution to the following data and test for goodness of fit at 5% level of significance. [15]

x:	0	1	2	3	4
f:	214	92	20	3	1

7. Cars arrive at a petrol pump with exponential interval times having mean $\frac{1}{2}$ minute. The attendant takes an average of $\frac{1}{5}$ minute per car to supply petrol and the service time being exponentially distributed. Find:
a) the average number of cars waiting to be served
b) the average number of cars in the system and
c) the proportion of time for which the pump attendant is idle. [5+5+5]

8. A gambler has Rs. 2. He bets 1 at a time and wins Rs. 1 with probability $\frac{1}{2}$. He stops playing if loses Rs. 2 or wins Rs. 4. Find:
a) the tpm of the related Markov chain
b) the probability that he has lost his money at the end of 5 plays.
c) the probability that the game lasts more than 7 plays. [5+5+5]

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